- **PERIYAR RIVER**
- It is famous for the Mulla-Periyar dam. It is a dam disputed between Kerala and Tamilnadu
- **PAMBA RIVER**
- It is associated with the Sabarimala temple. It flows nearby Sabarimala temple

CLIMATE OF INDIA (07:39 PM)

- **Factors affecting the Climate of India**
- a) Latitudinal Extent- The difference between the Southernmost and Northernmost latitudes which is 30 degrees. The overall climatic condition of tropical monsoons varies at the local levels due to the large latitudinal extent.
- b) Location- At the north of the Indian Ocean with a major part of the peninsular region surrounded by ocean resulting moderating effect.
- c) Presence of the Himalayas- It protects India from the cold and dry winds of Central Asia during winters. It also influences monsoons in India.
- d) Other topographical features- Such as the Western Ghats being perpendicular to Monsoon winds receiving high precipitation on the western side and rainshadow along the eastern side. The orientation of Aravali parallel to Monsoon winds resulting very low precipitation in Rajasthan
- e) Monsoon winds- The most dominant factor of Indian climate due to which it is called as Monsoon climate.
- f) Jet stream- Subtropical westerly jetstreams, Tropical easterly jetstreams, and Somali Jetstreams influence winter and summer conditions over India.
- g) Cyclones- The coastal regions of Tamilnadu, Andhra Pradesh, Orissa, and West Bengal are largely affected by Tropical cyclones.
- h) El-Nino, La-Nina, Maden-Julian oscillation, Indian Ocean Dipole, etc influence rainfall over India.

MONSOON (07:53 PM)

- The word Monsoon is derived from the Arabic word "Mausim" which means a seasonal reversal of wind.
- **Classical theory of the origin**
- This explains the phenomenon of Monsoons as large-scale Sea breezes and Land breezes due to the reversal of temperature and pressure conditions from summer to winter in the Northern Plains.
- During summers, high temperatures and low pressure attract winds from the Arabian Sea.
- During winters, Low temperatures and high pressures cause winds to blow away from the plains towards the sea.

The topic for the next class:- The Modern Theory of Monsoon

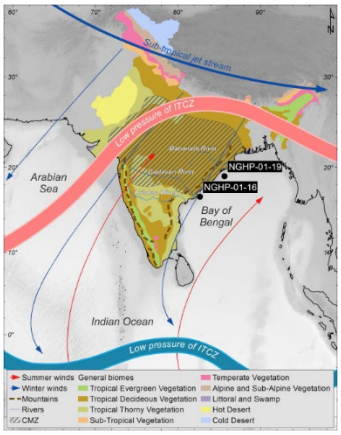
Geography Class 40

A BRIEF OVERVIEW OF THE CLASS (05:12 PM)

MODERN THEORY OF MONSOON (05:24 PM)

- **Winter season (November- February)**
- Low temperature and high-pressure conditions are prevailing over the northern plains.
- It causes Anti-cyclonic circulations causing winds to blow outward from the land over which air is subsiding.
- The sky is clear and dry.

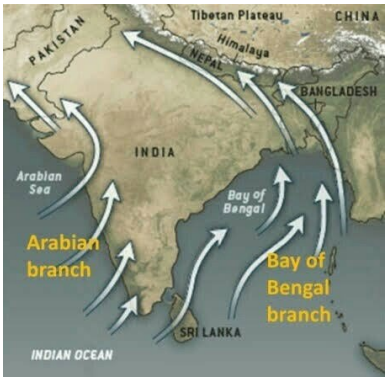
- The subtropical westerly Jetstream is strong and well-established. It is bifurcated into two branches i.e. towards the north and south of the Tibetan plateau. This southern branch is stronger and is well-established over the northern Ganga plains.
- This intensifies surface Anti-cyclones and causes winds to blow from land to sea in a North-east direction.
- North-east winds after crossing the Bay of Bengal pick up moisture and causes precipitation along the Coromandel coast.
- **Spring season (March-May)**
- The temperature has started to rise and pressure is falling
- The weather is extremely hot and dry
- The subtropical westerly Jet stream begins to weaken and the southern Branch eventually moves entirely to the North of the Tibetan Plateau
- This triggers convection and precipitation in some regions. They are called Pre-monsoon showers.
- Mango showers in Kerala- It helps in the ripening of mangoes
- Blossom showers in Karnataka- It helps in coffee flowering
- Norwesters or Kal Baishaki. In Assam, it is called as Bardoli Chheerha. It helps in Jute and rice cultivation
- Loo in northern plains from Punjab to Bihar.
- **Summer season (May-August) (06:26 PM)**
- The weather is extremely hot and dry. High temperature causes the complete development of low-pressure cells over deserts and the Northern Plains
- ITCZ gradually shifts toward the North and attracts the southern trade winds toward the north of the equator.
- These southern Trade winds after crossing the equator turn right under the influence of Coriolis force and starts to blow as southwest Monsoon winds.
- Temperature stratification in the air will not allow large-scale convection preventing major storms.
- The tropical easterly jetstreams and depressions in the Bay of Bengal help in disturbing the stratified air and triggers large-scale precipitation.
- The tropical easterly Jet stream is a low-level Jet stream existing over South Asia and Africa in Summer. It causes a change of divergence to convergence on the surface.
- The sudden onset of moisture-laden winds associated with violent Thunder and lightning is called as Bursting of the Monsoon.
- The Somali jetstream is a local jetstream off the coast of East Africa. It strengthens High pressure near Madagascar. The High-pressure cell causes a faster flow of winds toward India.
- In some regions due to local stability conditions, temperature stratification reappears or the wind starts blowing parallel to the Topography resulting in Break in the Monsoon.

- 

- **Autumn season (September- October)**
- ITCZ or Monsoon trough starts to move back gradually towards the south.
- This also brings back the maximum extent up to which southwest winds blow which is called the retreat of Monsoon.
- The southwest winds slowly get replaced by North-east winds.
- The high temperature with dry conditions existing over the Northern plains during October is called October Heat.
- The subtropical westerly Jet streams reappear to the south of the Tibetan Plateau creating dry conditions.

CHARACTERISTICS OF INDIAN MONSOON (07:08 PM)

- The onset of monsoon is gradual and withdrawal is more gradual.
- As the wind moves from sea to land, the amount of precipitation gradually decreases.
- The duration of the monsoon also decreases from sea to land.
- Temporal variation- Variation of rainfall with time.
- Spatial variation- Variation of rainfall from one location to another.
- **Distribution of Monsoon rainfall**
- The monsoon winds get divided into two branches- The Arabian Sea branch and the Bay of Bengal branch
- **<u>Bay of Bengal Branch</u>**
- It moves parallel to the Coromandel coast and causes no precipitation. From the K-G delta, rainfall starts to increase towards West Bengal.
- Along the Ganga delta, the Bay of Bengal branch is divided into eastern and western by Himalayan Foothills.
- The Eastern branch hits Meghalaya Plateau perpendicularly causing very heavy precipitation.
- The Western Branch moves across the Ganga plains.
- The precipitation decreases from east to west along the Ganga plains
- Precipitation increases towards the Himalayas from South to North.

- 

- **Arabian Sea Branch**
- It hits the Western Ghat at the right angle causing heavy precipitation along the western side.
- Along the Eastern side, the descending winds result in a rainshadow effect creating dry conditions in Maharashtra and Karnataka.
- The Arabian Sea Branch advances towards Gujarat and blows parallel to Aravallis resulting in very less precipitation in Rajasthan.
- The Arabian Sea Branch meets the Bay of Bengal Branch near Agra and the combined branch causes precipitation in the regions to the North.

WESTERN DISTURBANCES (07:46 PM)

- Western Disturbance is active between November to April.
- During winters Northern India experiences cold and dry conditions along with strong westerly Jet streams.
- Mediterranean region receives rainfall due to onshore westerlies and temperate cyclones.
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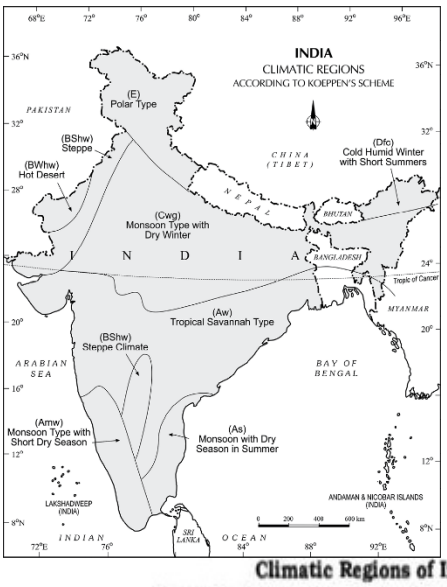
- The subtropical westerly Jetstreams along the Mediterranean region pick up the moisture content and low-pressure disturbance and are brought to India.
- When these winds accumulate near North-western Himalayas it causes precipitation.
- The precipitation due to western disturbance decreases from west to east along the Ganga plains
- **Significance of Western disturbances**
- It causes an abrupt decrease in temperature and snowfall along the northern hills
- It may result in cloudbursts and flash floods. example- 2010 Leh Cloudburst
- It is good for winter crops of wheat and mustard.
- **Impact of El-Nino, La-Nina, Indian Ocean Dipole, and Maden Julian Oscillation on Indian Monsoon.**
- Refer to oceanography.

CLIMATIC REGIONS OF INDIA (08:04 PM)

- **Refer to PPT in the class**
- 

- **Climatic Regions of India According to Koeppen's Scheme**

Type of Climate	*Areas*
Amw Monsoon with short dry season	West coast of India south of Goa
As – Monsoon with dry summer	Coromandel coast of Tamil Nadu
Aw – Tripical savannah	Most of the Peninsular plateaus, south of the Tropic of Cancer
Bwhw – Semi-arid steppe climate	North-western Gujarat, some parts of western Rajasthan and Punjab
Bwhw – Hot desert	Extreme western Rajasthan
Cwg – Monsoon with dry winter	Ganga plain, eastern Rajasthan, northern Madhya Pradesh, most of North-east India
Dfc – Cold humid winter with short summer	Arunachal Pradesh
E – Polar type	Jammu and Kashmir, Himachal Pradesh and Uttaranchal

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The topic for the next class:- Soils.
Geography Class 41

A BRIEF OVERVIEW OF THE PREVIOUS CLASS AND Q&A SESSION (05:09 PM)
SOILS OF INDIA (05:23 PM)

- **Some common characteristics of Indian soils**
- Nitrogen- All Indian soils are poor in Nitrogen.
- Laterite soil and Red soil are rich in Iron.
- Humus content is rich in Black soil, Mountain & forest soil, Peaty soil
- Salts- In Arid soil
- Black soil is black because the **titaniferous magnetite** compound
- **RED SOIL**
- Weather conditions- Moderate precipitation and Moderate to high temperature
- Parent rock- Granite and Gneiss
- It is rich in Iron, magnesium, and Aluminum and Poor in nitrogen, phosphorous, and Humus content.
- Regions- North-east, Southern Karnataka, Southern Andhra Pradesh, Eastern Madhya Pradesh, Chattisgarh, Andhra Pradesh, Orissa, and Parts of Tamilnadu.
- **LATERITE SOIL**
- Weather conditions- High temperature and high precipitation.
- Heavy leaching causes Laterization
- It is rich in Iron, Aluminium, and Potash but deficient in Silica, Bases, humus, etc
- It is slightly acidic in nature.